STACKS, MODULI OF CURVES: LEMMA-CONTRACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2864-2874 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 2876-2897, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_moduli-curves.json (pb=1137, stmt 2864-2874, proof 2876-2897).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-contract' also exists in stacks-models — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E6F: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 471–476. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7L: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2897–2897. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme and $s \in S$ a point. Let $f : X \to S$ and $g : Y \to S$ be families of curves. Let $c : X \to Y$ be a morphism over S. If $c_{s,*}\mathcal{O}_{X_s} = \mathcal{O}_{Y_s}$ and $R^1 c_{s,*}\mathcal{O}_{X_s} = 0$, then after replacing S by an open neighbourhood of s we have $\mathcal{O}_Y = c_*\mathcal{O}_X$ and $R^1 c_*\mathcal{O}_X = 0$ and this remains true after base change by any morphism $S' \to S$.*

Proof. Let $(U, u) \to (S, s)$ be an étale neighbourhood such that $\mathcal{O}_{Y_U} = (X_U \to Y_U)_*\mathcal{O}_{X_U}$ and $R^1(X_U \to Y_U)_*\mathcal{O}_{X_U} = 0$ and the same is true after base change by $U' \to U$. Then we replace S by the open image of $U \to S$. Given $S' \to S$ we set $U' = U \times_S S'$ and we obtain étale coverings $\{U' \to S'\}$ and $\{Y_{U'} \to Y_{S'}\}$. Thus the truth of the statement for the base change of c by $S' \to S$ follows from the truth of the statement for the base change of $X_U \to Y_U$ by $U' \to U$. In other words, the question is local in the étale topology on S. Thus by Lemma 0E6F we may assume

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X and Y are schemes. By More on Morphisms, Lemma 0E7L there exists an open subscheme $V \subset Y$ containing Y_s such that $c_*\mathcal{O}_X|_V = \mathcal{O}_V$ and $R^1c_*\mathcal{O}_X|_V = 0$ and such that this remains true after any base change by $S' \to S$. Since $g : Y \to S$ is proper, we can find an open neighbourhood $U \subset S$ of s such that $g^{-1}(U) \subset V$. Then U works. $\square$